

```
1 import numpy as np
2 population = np.array([45, 50, 55, 60, 65, 70, 75, 80, 85, 90])
3
4 # Random sampling
5 sample = np.random.choice(population, size=5, replace=False)
6
7 # Mean estimation
8 mean_estimate = np.mean(sample)
9
10 # Variance estimation
11 variance_estimate = np.var(sample)
12
13 print("Population Data:", population)
14 print("Sample Data:", sample)
15 print("Estimated Mean:", mean_estimate)
16 print("Estimated Variance:", variance_estimate)
```

Population Data: [45 50 55 60 65 70 75 80 85 90]

Sample Data: [65 55 75 90 50]

Estimated Mean: 67.0

Estimated Variance: 206.0